Conjoint Analysis

What is Conjoint Analysis?

- Marketing research technique originated in mathematical psychology and was developed by marketing professor Paul Green at the Wharton School of the University of Pennsylvania.

- Used to understand customer prefers components of product and service.

- This helps in designing new products or services that better meet customers' underlying needs. Conjoint analysis has been found to be an extremely powerful of way of capturing what really drives customers to buy one product over another and what customers really value.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL

What is Conjoint Analysis?

- Conjoint means “joined together”
- In Conjoint Analysis, features or attributes are considered jointly
- Conjoint analysis and trade-off studies are amongst the most sophisticated forms of market research because they provide estimates of demand based on product or service design.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL

Objective

- **Conjoint analysis** is a statistical technique used in market research to determine how people value different features that make up an individual product or service. The objective of conjoint analysis is to determine what combination of a limited number of attributes is most influential on respondent choice or decision making. A controlled set of potential products or services is shown to respondents and by analyzing how they make preferences between these products or services, the implicit valuation of the individual elements making up the product or service can be determined. These implicit valuations (utilities or part-worths) can be used to create market models that estimate market share, revenue and even profitability of new designs

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL

Fundamentals

- Conjoint analysis has its earliest roots in psychology and the testing of the theory that when people take decisions the result is the 'sum' of all the bits of value for each part in that decision. So when we buy a computer, there is a difference in value between computers with a small screen compared to a large screen, for instance. The difference in value to the customer can be linked just to this one feature and by adding up different values you can come to a conclusion about the overall value of the product or service.
- This seemed sensible in theory, but to test it required a method for breaking a product down into constituent parts, building profiles from these parts, then gathering preference data, then finally testing untried combinations to see if the customer preference was as per expectation. This is then the root of conjoint design.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL

Methodology

- Find main attributes and levels to build several product concepts
- Respondents were asked to rate/rank those product concepts
- Analyzing the data using suitable statistical techniques we derive the utility of each of the features

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL

Realistic Method

- Question:

  Which bundle do you prefer in your car?

- 1500 cc engine - 110 HP - 12 KMPL	OR	- 1000 cc engine - 80 HP - 20 KMPL

- Realistic trade off scenarios
- When respondents are asked to opt from difficult choices researcher can find out what they really value

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL

Model Workflow

- Inputs:
- Attributes
- Levels
- Respondents
- Expert Knowledge
- Research
- Experimental Design

- Outputs:
- Utility Scores for different levels
- Importance Scores for different attributes
- Ability to perform simulations

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL

Attributes

- Attributes form the fundamental basis of conjoint analysis. The idea is that a product or service can be broken down into its constituent parts (independent) - so for instance a mobile phone has a size, weight, battery life, size of address book, type of ring. Each of these elements making up a generic mobile phone is known as an attribute.

- No. of attributes
- Attributes must be independent

(Brand and Quality; Space and number of seats are similar attributes)

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL

Levels

- When we compare between mobile phones each will have a different specification on each of these attributes. We might have choices in terms of battery life between 12, 24, 36, 48 hours of battery life. Each of these options is known as a level of the battery life attribute. In another example, a car might have an attribute colour which is made up of the levels red, blue, green, white.

- In addition, each level can also be thought of as a performance target. If the customer, on the attribute 'delivery', wants the level "next day delivery" and we are perceived to be offering "48hr delivery" then there is a gap to make up. What is more with conjoint analysis, it is possible to calculate what the value of making up that gap is to the business. Compare this to the cost of doing it and the business can decide whether it is worth it.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL

General rules

- Don’t include many attributes or levels
- Levels and attributes should freely go into making a possible product, there shouldn’t be any impossible combination as such
- Each level also needs to be capable of being read and understood on its own. Although attributes are used to help break a product down and in analysis, when presented to respondents what respondent sees is the levels.
- For many products, particularly in business markets, service can be more important than the actual product. By using both product and service attributes in the same conjoint it is possible to see how customers trade-off service against features. The care, however, has to be taken to balance the attributes to prevent biasing the outcome one way or another.
- It is also vital that the levels used describe real and where possible measurable and actionable performance.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL

Utilities (Part Worths)

- The numbers tell you how preferable different features are:

Ice Cream Flavor	Utility
• Vanilla	2.5
• Chocolate	1.8

Ice Cream Price	Utility
• Rs 25	5.3
• Rs 35	3.2
• Rs 50	1.4

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL

Simulation Exercise

- Derive preference between Rs 35 Vanilla Cone vs Rs 25 Chocolate Cone for a new customer

- Vanilla (2.5) + Rs 35 (3.2) = 5.7
- Chocolate (1.8) + Rs 25 (5.3) = 7.1

- The new customer chooses Rs 25 chocolate cone

Generate Orthogonal Design

- A major hurdle was in reducing the number of profiles down to a manageable number. This drew on work from statistics looking at experimental design - how do you minimize the number of experiments required to test a number of combinations of properties, for instance in drug development? The result was the use of 'orthogonal fractional factorial plans' in building the profiles for testing. Since the objective was to validate the decision making process, tests were carried out and analyzed and then compared to 'hold-out' profiles. These were profiles that were not counted for statistical accuracy but were required to validate the model.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL

Generate Orthogonal Design

- The statistical technique of Fractional Factorial Design of Experiment finds out the minimum number of product designs which are necessary to use in the study and yet provide us all the information that we originally sought. These designs are also mutually independent (orthogonal) to avoid any redundancy in the data and allow the representation of each of the attributes and their respective levels in an unbiased manner.
- Note that such reduction in number of product designs is possible only after making certain assumptions. For example, we may assume that none of the attributes interact among themselves. Or in other words, the attributes are considered to be independent of each other.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL

A GENERAL PRESENTATION OF FULL PROFILE CONJOINT ANALYSIS

A complete CA includes several phases:

- Definition of a model that links product features to customer judgements. Definition of the profiles to submit to consumer judgement.
- A survey to collect consumer preferences.
- A statistical analysis aimed at estimating the so-called part-worth utilities and attribute importance.
- A simulation phase where market shares for submitted and new profiles may be estimated.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

Model

- Consumer preferences are decomposed according to an additive utility model, specific to each respondent. This is a very important feature of CA: there is no average or typical consumer, but different consumers who have different ways of weighting attributes and their categories.
- Suppose that a product is defined by a combination of levels (*i*, *j*,*k*,...) of *P* attributes: its global utility for a customer is equal to $ai + bj + ck +$ ..., a sum of part-worth utilities.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

Choice of the Profiles

Each profile is defined by a combination of a (generally small) number P of attributes with $m_1,\ldots,m_P$ categories. Most of the time, attributes are qualitative variables with few ordered or non-ordered categories, but may be also discretized continuous variables (e.g. price) with few levels. Proposed profiles should not include the 'perfect' product (e.g. a car with high speed, high comfort, high security and low price), since it would be trivially chosen by all consumers. On the contrary, CA assumes that each consumer makes a trade off between attributes by putting into balance advantages and inconveniences.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

Since the total number of products $\prod_{p=1}^{P} m_p$ to be compared is generally large, the subsets have to be carefully chosen

Paint X lite

- This is usually achieved through orthog- onal designs, which have optimal properties. In a balanced orthogonal design, all combinations of levels of each pair of attributes should be observed the same number of times.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

Collecting Consumer Preferences

- The basic experience consists in asking *Q* consumers to fully rank the fictitious products.
- Sorting products is a rather difficult task when there are many products to rank. So why not to rate products by giving scores to each of them?
- Rating products is much easier but has severe drawbacks:

1. scales across respondents may not be comparable
2. there is a risk of ties

- Rankings are thus usually preferred.
  It frequently happens that respondents are unable to give a complete order, but are able to rank only their 'best' products, the other products becoming ex aequo.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

Estimation of the partworth utilities

- Conjoint Analysis is used to determine partial utilities (“partworths”) β for all factor values based upon the ranked data
- Furthermore, with this partworths it is possible to compute the metric total utilities *y* of all incentives/profiles and the relative importance of the single object attributes

Let X be the binary design matrix with N rows and $\sum_{p=1}^{P} m_p + 1$ columns

- The *Q* ranks defined by consumer preferences are juxtaposed in

the response matrix *Y*.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

Estimation Result

- Ordinary Least Squares (OLS) regression of *Y* on *X* is commonly used for estimating coefficients assigned to categories, which are interpreted as part-worth utilities.

- The result of the analysis was the calculation of 'part-worths'. That is the model betas which describe how much each variable contributes to the final model. A higher beta is being more important. Since these part-worths have no units and because they are predicting an abstract entity such as preference, it is possible to scale and adjust the parameters without affecting the underlying model outcomes.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

Analysis and Beyond

Since for any respondent *q* (*q* = 1...*Q*), we know the *m*1 +⋯+*mP* part-worth utilities, it is possible to:

1. *Simulate new profiles*:
2. *Estimate market shares*
3. *Study consumer real purchase intents*

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

Case Study

In Conjoint Analysis the profile of different products are presented to the consumers for their responses. These profiles are generated by varying the levels of its attribute. For example, suppose we are conducting a Conjoint Analysis based study of dish washers. Let us assume that the most important attributes considered by its customers are Brand, Price, Washing Capacity, Colour and Shape. Let us further assume that the following levels of attributes are considered relevant and interesting by the marketer for the study.

S.No.	**Attribute**	**Levels**
1.	Brand	Kitchen Master; Elegant; Torrent Evermaid
2.	Price	Rs. 15,000/-; Rs. 20,000/-; Rs. 25,000/-
3	Washing Capacity	High; Medium; Low
4.	Colour	Steel grey; White; Light Blue; Pink
5.	Shape	Cylindrical; Box Type

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

Since the 5 attributes can take 4, 3, 3, 4 and 2 levels, the total number of possible product concepts that can be generated by configuring these attributes is $4\times3\times3\times4\times2=288$. In order to determine the part worth utilities of each of the levels of all these attributes, we shall have to take 288 different product concepts for getting his/her responses. This number is certainly too large for any consumer. Therefore, we resort to the method of Fractional Factorial Design of Experiment to make it manageable.

The statistical technique of Fractional Factorial Design of Experiment finds out the minimum number of product designs which are necessary to use in the study and yet provide us all the information that we originally sought. These designs are also mutually independent (orthogonal) to avoid any redundancy in the data and allow the representation of each of the attributes and their respective levels in an unbiased manner.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

- In the example of dish washer considered here, this technique has given us only 16 designs out of the 288 possible dish washers. The 16 product concepts found through this method are not unique. Many other sets of 16 cards would have also been equally good. However, all of these sets would have to be independent and represent all the attributes and their respective levels in an unbiased manner. We are illustrating below one such set of 16 cards representing the product concepts of dish washers using Fractional Factorial Design of Experiment.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

- Ease of data collection is a key feature of Conjoint Analysis. The consumers are asked only to assign rating scores to each of the product stimuli or even rank the different concepts presented to them. This is quite a realistic task and is close to the shopping experiences where the customer merely makes choices. He does not have to respond to each of the attributes separately.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

Estimation of Part-worth Utilities

- After using any of the available method, the output is obtained for each of the respondent separately. This is quite significant as the disaggregate data can be combined in any of the desired way. But, if the output was only at the aggregate level then disaggregation might not have been possible.
- In our example of dish washer, the part worth utilities may be found for each of the attributes and their levels as following:

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

Table 2: Attributes with their levels

	Attribute	Levels	Part Worth Utility
1.	Brand (25.4%)	Kitchen Master	2.2
		Elegant	-2.3
		Torrent	-0.8
		Ever Maid	0.9
2.	Price (66.2%)	Rs. 15,000	5.0
		Rs.20,000	-0.1
		Rs. 25,000	-5.8
3.	Washing Capacity (4.9%)	High	0.0
		Medium	0.1
		Low	-0.1
4.	Colour (2.8%)	Steel Grey	-0.3
		White	-0.1
		Light Blue	0.1
		Pink	0.2
5.	Shape (0.7%)	Cylindrical	-0.1
		Box Type	0.1

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document

Analysis of the Result

- From the above table, we find that price plays most important role (66.2%) in the minds of customers. This is followed by Brand(25.4%), Washing Capacity(4.9%), Colour (2.8%) and Shape (0.7%). These relative importance values for the maximum and minimum values of the part, worth utilities of the respective attributes. Customers respond quite predictably towards different price levels. They prefer lesser price to the higher ones. Among brands, we find that Kitchen Master is the most liked brand and Elegant in the least liked one. However, they do not prefer the "High" capacity most.They prefer "Medium" size most. In terms of the colour, they prefer Pink colour most. Between the two shapes, box type dish washers are preferred more. But, the importance scores for the wash capacity, colour and shape themselves are quite low.
- The part worth utilities can now be added to determine the total utility for each of the possible product concepts. This allows us to scan the consumer preference pattern for all of the 288 product concepts although he has been exposed to only 16 of them. We can now also rank all of these 288 product concepts.

https://egyankosh.ac.in/bitstream/123456789/12634/1/unit-27.pdf, NPTEL, https://hal-cnam.archives-ouvertes.fr/hal-02941462/document